

Involvement-Aware Foundation Models for Prostate Cancer Detection in Ultrasound

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Introduction: Prostate cancer (PCa) detection from ultrasound (US) using deep learning (DL) models has shown potential for enhancing real-time guidance during biopsies. However, prostate US images lack pixel-level cancer annotations, and are instead labeled through coarse histopathology annotations of the entire core, introducing weak labels. Current ap-

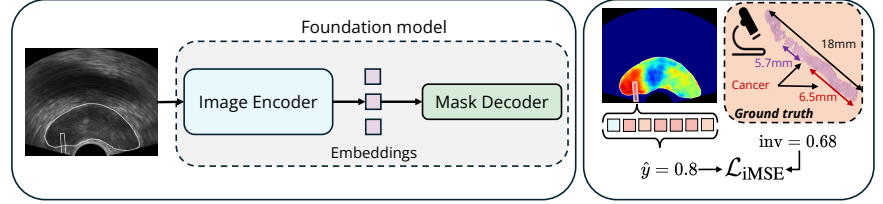


Figure 1: High-level overview of our training method

proaches [1, 2] to PCa detection fail to address this issue due to their focus on local features and lack of context awareness, with foundation models [3, 4] offering a promising alternative to these methods by analyzing entire images to capture global context. However, despite this improvement, foundation models still encounter challenges stemming from weak labels, leading to imprecise localization of cancer in the image. We propose to address this limitation by designing a specialized loss function that incorporates cancer involvement, defined by the proportion of cancer tissue reported by pathology, into the training objective. By providing the model with a more complex form of supervision, our proposed method improves the robustness of foundation models for PCa detection.

Methods: We use 2631 biopsy cores collected from 311 patients who underwent prostate biopsy in 2 centers under the guidance of trans-rectal ultrasound (TRUS). Our proposed method uses MedSAM [4], a foundation model for medical images, to segment cancer in the US image using involvement as a ground truth. We design the involvement-aware mean-squared error (iMSE) loss function to minimize the distance between the average intensity of the model’s predictions within the region of interest and the reported cancer involvement of the core. iMSE aims to mitigate the impact of coarse labeling by allowing the model to learn some information regarding the spread of cancer in the core. The model’s final prediction is computed by averaging the activations of the segmentation output within the labeled region, and iMSE is computed using: $\mathcal{L}_{\text{iMSE}}(\hat{Y}, \text{inv}) = (1/R) \sum_i \sum_j (\hat{Y}[i, j] \cdot \mathbb{1}_{\hat{Y}[i, j] \in R} - \text{inv})^2$.

Results: Our results are shown in Table 1. We compare our method against iLR [1], a strong baseline method that minimizes the difference between true and predicted involvement to combat label noise, and 2 foundation models, SAM [3] and MedSAM [4]. Our proposed method leads to superior cancer detection, outperforming all baselines, with an AUROC of 75.2 (+1.3%), a balanced accuracy of 69.3 (+0.4%), and better sensitivities across all thresholds (+1.7%, +3.9%).

Method	AUROC	Bal. Acc.	Sen.@40Spe	Sen.@60Spe
iLR [1]	67.0	63.7	83.1	67.0
SAM [3]	71.2	68.4	80.6	68.4
MedSAM [4]	73.9	68.9	81.9	71.7
MedSAM +iMSE	75.2	69.3	85.7	75.6

Table 1: Comparing the performance of our method against prior work and baselines for PCa detection

Conclusion: Our findings underscore the promise of involvement-aware loss functions in advancing foundation models for weakly labeled ultrasound data. Future work can further enhance robustness by training models to learn variations of prostate cancer tissue over time.

References

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